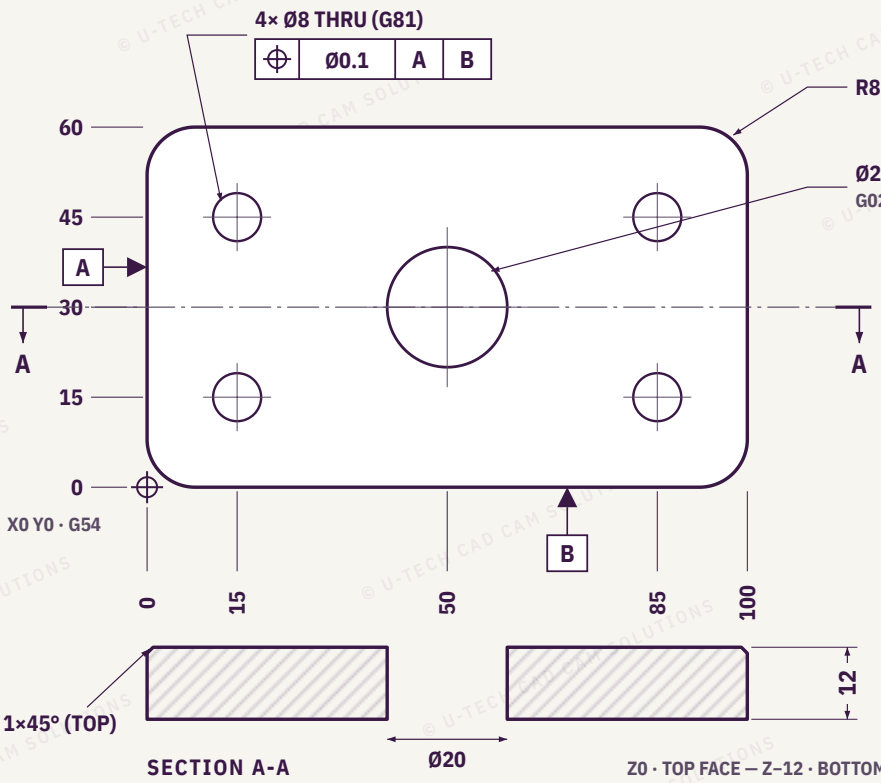
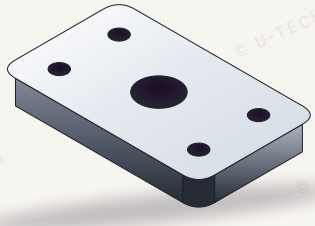
VIEW 01 PLAN VIEW · SECTION A-A — Ordinate dims from G54 datum · mm



VIEW 02 ISOMETRIC
Rendered · not to scale



SETUP · G54

X0 Y0 — bottom-left corner of plate
Z0 — top face · stock 105 × 65 × 15 pre-faced
Work holding — vice on parallels, stop on left
Start block — G90 G54 G17 G40 G80 G49

ASSIGNMENT 1 · BASIC · OBJECTIVES

1. **Safe start, tool change M06, G43 length offset**
2. **Outer contour with G41 cutter compensation**
3. **G81 drilling cycle with R-plane, G80 cancel**
4. **Ø20 hole by G02/G03 circular interpolation**
5. **Submit: program + setup sheet + inspection report**

GENERAL NOTES

01 Material: AL 6061-T6, stock 105 × 65 × 15.
02 All dimensions in mm, from datum X0 Y0.
03 Untoleranced dimensions: ISO 2768-mK.
04 Break edges 0.2×45°; 1×45° on top edge only.
05 Contour: G41 climb, Ø10 EM, one pass, Z–12.5.
06 Ø8 holes: spot, then G81 Z–14 R2 F100.
07 Ø20: Ø10 EM, G03 full circle with I/J, 3 depths.
08 Feeds: S4000 F600 mill · S2500 F100 drill.

STUDENT _____ ROLL NO. _____ DATE _____ MARKS / 20 _____

FEATURE TABLE

G54 · mm

ID	X	Y	FEATURE	Z
H1	15	15	Ø8 H12	THRU
H2	85	15	Ø8 H12	THRU
H3	15	45	Ø8 H12	THRU
H4	85	45	Ø8 H12	THRU
H5	50	30	Ø20 H9 (G03)	THRU
C1	—	—	CONTOUR R8	–12.5

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TITLE

DRILL PLATE

ASSIGNMENT 1 · BASIC

DRG. NO. UTCCS-VMC-A1-001	SHEET 1 / 1
MATERIAL AL 6061-T6	STOCK 105 × 65 × 15
SCALE NTS · UNIT mm	PROJECTION 1st angle
GENERAL TOL. ISO 2768-mK	SURFACE Ra 3.2 (√)

LIMITS OF SIZE

FEATURE	FIT	UPPER	LOWER
Ø20	H9	+0.052	0.000
Ø8	H12	+0.150	0.000
THK 12	—	+0.050	–0.050

REQUIRED TOOLING

VMC · 3-AXIS

T01	Face top	Ø50 face mill
T02	Contour G41 · Ø20 G03	Ø10 end mill
T03	Spot all holes	Ø6 spot 90°
T04	Through holes G81	Ø8 drill

Gauges: Ø20 H9 plug · Ø8 plug · 0–25 micrometer · vernier

Marking: program 8 · setup 4 · part 6 · report 2

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